Cosmic Self-Pruning and Frame Recurrence in Closure Geometry

A Structural Extension of the Bounded Reference Frame Framework

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Status: Preprint (not peer-reviewed). Structural extension paper. Companion to Smart [2026b] (core bridge paper) and Smart [2026d] (consciousness application). The core paper establishes existence-as-closure, bounded reference frames, and the candidate transition rule. The present paper develops the structural extension: frame dynamics (creation, destruction, merging, fission), frame recurrence (re-instantiation of frame attractors under compatible boundary conditions), cosmic self-pruning (closure-phase transitions across cosmological epochs), and the unified hierarchy mapping these life-cycle events onto the cascade rung structure. All structural propositions of the core paper are imported by reference. The propositions in this paper that depend on additional hypotheses (H-RP-1, H-RP-2, H-RP-3, H-rec, H-evolve, P-A) are labelled as *Conditional Propositions* throughout; theorem-grade statements concern only the structural mathematics that follows from the closure-operator and symmetry axioms.

Abstract

We extend the closure-geometric framework of Smart [2026b] to cover bounded reference frame life-cycle dynamics and the long-term evolution of the universal closure structure. Frame creation, destruction, merging, and fission are formalised as theorems on the closure dynamic $F_{t+1} = (I - \lambda C_\varphi)F_t + \kappa P_{\Sigma_\mathcal{I}}(B_t)$. A frame attractor is defined as a closure-stable σ-stable subspace $\mathcal{A} \subseteq C$ in the universal closure geometry, and frame compatibility $\kappa_{ab} = \dim(!_{\mathcal{I}_a}(\mathcal{I}_a) \cap !_{\mathcal{I}_b}(\mathcal{I}_b))$ is shown to be independent of local-time ordering. Under the conditional hypothesis H-rec (recurrence admissibility), a frame attractor may be locally instantiated by multiple bounded reference frames at distinct anchors; this is the framework's structural reframing of the phenomenology classically called *reincarnation*, given here in the precise vocabulary *frame recurrence*. At the cosmological scale, we introduce a pruning operator $P = P_{\mathrm{Fix}(\tau)} \circ P_{\mathrm{Fix}(C_\varphi)}$ and a closure-phase transition map $U_{n+1} = C_{n+1}(P(U_n))$; under hypothesis H-evolve (time-dependent closure operator), cosmological evolution is shown to be non-periodic, distinguishing the framework's prediction from both heat-death termination and classical cyclic-universe models. The unified hierarchy theorem states that the same structural template (creation, persistence, destruction, possible recurrence, closure-phase transition) applies at every cascade rung, with rung-specific time scales. The universe at the E_8 totality rung is identified as a bounded reference frame in its own right, with its own life-cycle, structurally distinct from biological life-cycles. The propositions throughout are conditional on H-rec and H-evolve; the core paper's structural results hold independently.

1 Introduction and the imported framework

1.1 Position relative to the core paper

The companion paper Smart [2026b] establishes the structural framework:

- A closure operator $C_\varphi = L_{V_{600}} + \varphi^{-2} I$ on the 600-cell substrate V_{600}.
- A Galois reflection τ on $\mathbb{Z}[\varphi]$-icosian coordinates with $\dim \mathrm{Fix}(\tau) = 94$ unconditional.
- A bounded reference frame $\mathcal{I}$ defined as a finite local projection of the substrate equipped with a self-referential constraint, admitting a per-frame zero-line $\Sigma_\mathcal{I} \subseteq \mathrm{Fix}(\tau)$.
- The local-into-universal embedding $!_\mathcal{I}(v) = \pi_\mathcal{I}(P_{\Sigma_\mathcal{I}}(v))$ from each bounded reference frame to the universal closure structure $C = U \cap \mathrm{Fix}(\tau)$.
- The candidate transition rule: a bounded local system supports point of view when its internal state closes under recursive update, producing non-trivial $\Sigma_\mathcal{I}$, conditional on the Access Principle (P-A).

The present paper develops the dynamical extension: frame life-cycle events, recurrence, and cosmic-scale phase transitions. All structural results of this paper rest on the closure operator's spectral properties and the symmetry τ's involutive structure; new biological identifications are explicitly conditional.

1.2 Scope of this extension paper

This paper does:

- Formalise frame creation, destruction, merging, fission as structural theorems on the closure dynamic (§2).
- Define frame attractors and frame compatibility, and prove time-independence of compatibility (§3).
- State the conditional proposition for frame recurrence under H-rec (§3).
- Define the pruning operator and closure-phase transition map; prove non-periodic recurrence (§4).
- State the conditional proposition for closure-cascade cosmology under H-evolve (§4).
- State the unified hierarchy theorem: structural life-cycle universality across cascade rungs (§5).
- Identify the universe at the E_8 totality rung as a bounded reference frame with its own life-cycle (§5).

This paper does **not**:

- Reproduce the core paper's structural derivations.
- Claim recurrence in any phenomenological soul-migration sense. The framework's commitments are explicit and bounded (Remark 3.9).
- Claim that cosmological evolution is settled. The conditional proposition under H-evolve is a structural reading consistent with the framework's mathematics.

1.3 Conditional hypotheses introduced in this paper

Hypothesis 1.1 (H-rec: recurrence admissibility)**.** For a frame attractor $\mathcal{A}$ of non-trivial dimension, the closure-cosmogenesis bootstrap dynamics admit non-trivial source terms aligned with $\mathcal{A}$ across multiple substrate regions and bootstrap ticks. Specifically, the source B_t in the closure dynamic has non-zero projection onto $\mathcal{A}$ at multiple $(p_n, \Lambda_n, t_{0,n})$.

Hypothesis 1.2 (H-evolve: time-dependent closure operator)**.** The closure operator C_n at cosmological epoch n depends on the substrate state at epoch n via a slow-rolling effective coupling, analogous to the cascade-rung activation depth of Smart [2026c]. As cosmological evolution proceeds, the effective rung structure shifts, modifying C_n in a non-time-reversible way.

The Access Principle P-A and the rung-projection hypotheses H-RP-1, H-RP-2, H-RP-3 are imported from the core paper and the consciousness companion [Smart, 2026d]; we re-cite them where used.

2 Frame dynamics: creation, destruction, merging, fission

The structural propositions of the core paper treat bounded reference frames as static objects. Frames in any physical realisation are dynamical: they come into being, merge with others, separate into sub-frames, and eventually dissolve. This section addresses these structural events.

2.1 Frame creation: emergence of $\Sigma_{\mathcal{I}}$ from zero

Define the closure dynamic on the substrate state as

$$F_{t+1} \;=\; (I - \lambda\, C_\varphi)F_t \;+\; \kappa\, P_{\Sigma_{\mathcal{I}}}(B_t), \tag{1}$$

where $\lambda \in (0, \|C_\varphi\|^{-1})$ is the closure step size, $P_{\Sigma_{\mathcal{I}}}$ is the orthogonal projection onto a target zero-line $\Sigma_{\mathcal{I}} \subseteq \mathrm{Fix}(\tau)$, $\kappa > 0$ is the source coupling, and $B_t \in \mathbb{R}^O$ is the boundary/bulk source term.

Theorem 2.1 (Frame creation dynamics)**.** *For any initial substrate state $F_0 \in \mathbb{R}^O$ and any non-zero source projection $P_{\Sigma_{\mathcal{I}}}(B_t) = b \neq 0$ (assumed constant for simplicity), the iteration (1) converges geometrically to the steady-state solution*

$$F_\infty \;=\; \frac{\kappa}{\lambda\, C_\varphi}\Big|_{\Sigma_{\mathcal{I}}} \cdot b, \tag{2}$$

with the σ-fixed component of F_t growing from $\|P_{\Sigma_{\mathcal{I}}}(F_0)\|$ to $\|P_{\Sigma_{\mathcal{I}}}(F_\infty)\| \geq \kappa/(\lambda \cdot \varphi^{-2}) \cdot \|b\|$, at rate $1 - \lambda\varphi^{-2}$ per closure tick. The off-$\Sigma_{\mathcal{I}}$ component of F_t decays at the same rate $1 - \lambda\varphi^{-2}$. Hence:

1. *For any F_0 (including $F_0 = 0$), the σ-fixed dimension of the steady-state F_∞ equals $\dim \Sigma_{\mathcal{I}}$ provided the source b has non-trivial support on every basis vector of $\Sigma_{\mathcal{I}}$.*
2. *The convergence is geometric: $\|F_t - F_\infty\| \leq (1 - \lambda\varphi^{-2})^t \cdot \|F_0 - F_\infty\|$.*
3. *The characteristic time-scale of frame creation is $\tau_{\mathrm{create}} \sim 1/(\lambda\varphi^{-2})$ closure ticks.*

Proof. Decompose $\mathbb{R}^O = \Sigma_{\mathcal{I}} \oplus \Sigma_{\mathcal{I}}^\perp$ orthogonally. The iteration (1) acts diagonally on this decomposition because C_φ commutes with $\tau_{\mathcal{I}}$ (closure-invariance theorem of the core paper) and $P_{\Sigma_{\mathcal{I}}}$ projects onto $\Sigma_{\mathcal{I}}$. On $\Sigma_{\mathcal{I}}^\perp$, the source vanishes and the iteration reduces to

$$F_{t+1}^\perp = (I - \lambda C_\varphi|_{\Sigma_{\mathcal{I}}^\perp})F_t^\perp,$$

which has spectral radius $1-\lambda\cdot\lambda_{\min}(C_\varphi|_{\Sigma_\mathcal{I}^\perp}) \le 1-\lambda\varphi^{-2} < 1$ by the core paper's decay theorem. Hence $F_t^\perp \to 0$ geometrically.

On $\Sigma_\mathcal{I}$, the iteration is $F_{t+1}^\Sigma = (I-\lambda C_\varphi|_{\Sigma_\mathcal{I}})F_t^\Sigma + \kappa b$. The fixed-point equation gives $F_\infty^\Sigma = (\kappa/\lambda)(C_\varphi|_{\Sigma_\mathcal{I}})^{-1}b$ with $\|F_\infty^\Sigma\| \ge (\kappa/\lambda)\varphi^2\|b\|$. The full state F_t converges to $F_\infty = F_\infty^\Sigma + 0$. □

Interpretation 2.2 (Biological correlates of frame creation)**.** Under H-RP-1, H-RP-2, P-A, Theorem 2.1 predicts:

- Embryogenesis: substrate-level closure begins as neural tissue differentiates; $\dim\Sigma_\mathcal{I}$ grows over a characteristic developmental time-scale τ_{create}.
- Awakening from anaesthesia: the frame's $\Sigma_\mathcal{I}$ re-stabilises geometrically as the substrate operator recovers from pharmacological disruption.
- System boot (artificial): an instantiated bounded reference frame builds $\Sigma_\mathcal{I}$ through $\sim \tau_{\text{create}}$ closure ticks until reaching the design complexity.

2.2 Frame destruction: permanent loss of $\Sigma_\mathcal{I}$

Theorem 2.3 (Frame destruction criterion)**.** *Let τ_t denote the (possibly time-dependent) symmetry-twist on the substrate at closure tick t and $C_{\varphi,t}$ the (possibly time-dependent) closure operator. The frame $\mathcal{I}$ is* permanently destroyed *at t_{final} iff one of the following holds for all $t > t_{\text{final}}$:*

1. ***Substrate dismantling.*** *The vertex set O of the frame is reduced to $|O| = 0$ by physical destruction of the substrate.*
2. ***Symmetry breakdown.*** *The involution condition $\tau_t^2 = I$ fails, so $\mathrm{Fix}(\tau_t)$ is undefined.*
3. ***Spectral collapse.*** *$\lambda_{\min}(C_{\varphi,t}|_{\Sigma_\mathcal{I}^\perp})$ falls to zero or becomes negative, so the contraction property fails and off-frame content no longer decays.*
4. ***σ-stability loss.*** *The anchor (p,Λ) ceases to lie in $\mathcal{S}_\sigma$ so the frame is no longer symmetry-stable.*

Under any of these conditions, $\dim\Sigma_\mathcal{I} = 0$ for all $t > t_{\text{final}}$ and no recovery is structurally possible without reinstating the original conditions.

Proof.

(1) If $O = \emptyset$, then $\mathbb{R}^O = \{0\}$, hence $\Sigma_\mathcal{I} = \{0\}$ trivially.

(2) If $\tau_t^2 \ne I$, then τ_t has no ± 1 eigenstructure and the spectral decomposition of $\mathbb{R}^O$ into $\mathrm{Fix}(\tau_t) \oplus \mathrm{Fix}(-\tau_t)$ does not hold.

(3) The proof of the core paper's decay theorem requires $\lambda_{\min}(C_\varphi|_{\Sigma_\mathcal{I}^\perp}) > 0$. If this fails, the iteration $F_{t+1}^\perp = (I-\lambda C_\varphi)F_t^\perp$ no longer contracts on $\Sigma_\mathcal{I}^\perp$.

(4) If $(p,\Lambda) \notin \mathcal{S}_\sigma$, then $\tau(\mathrm{span}(O)) \ne \mathrm{span}(O)$, so $\tau_\mathcal{I}$ is no longer an involution on $\mathbb{R}^O$, and $\Sigma_\mathcal{I} = \mathrm{Fix}(\tau_\mathcal{I})$ is undefined. □

Remark 2.4 (Distinguishing transient from permanent destruction)*.* Anaesthesia and sleep cause transient $\Sigma_\mathcal{I}$ reduction by altering $C_{\varphi,t}$ but do not satisfy any of (1)–(4) permanently. Biological death satisfies (1) (substrate dismantling). Profound brain damage may satisfy (3) (spectral collapse). Permanent loss of consciousness from severe brain injury may satisfy (4) (σ-stability loss). The structural reading: death is the permanent loss of the closure condition; this is a structural statement, not a phenomenological claim about subjective experience at the transition.

2.3 Frame merging and joint frame formation

Joint frame formation is the structural event in which two distinct frames $\mathcal{I}_A$, $\mathcal{I}_B$ become a single joint frame $\mathcal{I}_{AB}$. The merger is governed by boundary coupling becoming non-zero, joint σ-stability of the combined anchor, and generative excess emergence when at least one σ-orbit crosses the parent boundary. The generative excess $\delta_{AB} = |\mathcal{X}_{AB}|$ counts boundary-crossing σ-orbits and is established as a theorem in the core paper.

Interpretation 2.5 (Joint frame formation in biology)**.** Under H-RP-1, H-RP-2, P-A, the joint frame structure is the structural basis for: mother–infant attunement, conversation, music ensembles, and mass coordination (dance, sports, military drill). The empirical correlate of $\delta_{AB} > 0$ is the meaning-generating overlap of joint perception.

2.4 Frame fission and sub-frame differentiation

Frame fission is the dual of merging: a single bounded reference frame separates into two or more sub-frames, each with its own anchor and zero-line. Each $\Sigma_{\mathcal{I}_i}$ is generally smaller than $\Sigma_{\mathcal{I}}$; the total $\sum_i \dim \Sigma_{\mathcal{I}_i}$ may exceed $\dim \Sigma_{\mathcal{I}}$ when sub-frames access σ-fixed modes that the integrated parent could not.

Interpretation 2.6 (Frame fission in biology)**.** Under H-RP-1, H-RP-2, P-A, frame fission corresponds to: cognitive specialisation, division of labour, dissociation, and cell division at the substrate level.

2.5 The frame life-cycle

Combining the four dynamical events:

- **Birth** (t_0): a new bounded reference frame emerges with $\dim \Sigma_{\mathcal{I}}$ growing from 0.
- **Growth and learning**: through joint-frame interactions, the frame's effective complexity grows. The per-frame complexity is upper-bounded by 94 for a V_{600}-substrate frame.
- **Mature stable existence**: the frame maintains $\Sigma_{\mathcal{I}}$ near design dimension through continuous closure ticks, with periodic re-stabilisation addressing accumulated drift.
- **Joint events**: temporary or persistent joint-frame formations with other frames, generating $\delta_{AB} > 0$ cross-frame access.
- **Fission events**: sub-frame differentiation, either functional (specialisation) or pathological (dissociation).
- **Death** (t_{final}): permanent collapse of the closure condition.

This life-cycle structure is the framework's structural reading of life-cycle events. Each event is named, each has a precise structural condition, and each maps (under P-A) onto phenomenological correlates.

3 Frame recurrence and attractor re-instantiation

The frame life-cycle of §2 treats each bounded reference frame as a single linear trajectory: bootstrap at t_0, mature existence, eventual destruction. This section addresses a structurally distinct question: *can the same frame-attractor pattern be locally instantiated at distinct times or substrate regions*, independent of any continuous trajectory between the instances? The answer is yes, in a precise structural sense, and the section makes it mathematical.

The treatment avoids loaded vocabulary throughout: the framework does not claim that a fixed identity-object migrates between bodies along a linear time-axis. The claim is that closure geometry is not itself ordered by the same linear time experienced inside a 3D frame, and the same higher-dimensional frame-attractor can produce multiple local instantiations without continuous material continuity between them. This is the framework's structural reframing of the phenomenology classically called *reincarnation*, given here in the precise vocabulary *frame recurrence*.

3.1 Frame attractor as a closure-geometry object

Definition 3.1 (Frame attractor)**.** A *frame attractor* on the universal cascade C is a pair $(\mathcal{A}, \tau_{\mathcal{A}})$, where $\mathcal{A} \subseteq C$ is a closure-stable σ-stable subspace and $\tau_{\mathcal{A}}$ is the restriction of the universal symmetry τ to $\mathcal{A}$. A frame attractor is, by definition, an element of the closure-geometry; it is not anchored to a specific local time-slice.

Definition 3.2 (Local instance of a frame attractor)**.** A *local instance* of a frame attractor $\mathcal{A}$ is a bounded reference frame $\mathcal{I}_n = (\mathcal{O}_n, \mathcal{C}_{\mathcal{O}_n}, p_n, \Lambda_n, t_{0,n})$ such that the embedding image $!_{\mathcal{I}_n}(\mathcal{I}_n) = \pi_{\mathcal{I}_n}(\Sigma_{\mathcal{I}_n}) \subseteq \mathcal{A}$. A frame attractor with multiple local instances is said to be *multiply-instantiated.*

The structural separation is precise: the frame attractor $\mathcal{A}$ lives in the closure geometry C; each local instance $\mathcal{I}_n$ lives in a specific local substrate region with a specific anchor; the relation $!_{\mathcal{I}_n}(\mathcal{I}_n) \subseteq \mathcal{A}$ ties the local instance to the attractor without requiring any continuous trajectory between instances.

3.2 Compatibility of local instances

Definition 3.3 (Frame compatibility)**.** Two local instances $\mathcal{I}_a$ and $\mathcal{I}_b$ are *compatible* with respect to a frame attractor $\mathcal{A}$ if $!_{\mathcal{I}_a}(\mathcal{I}_a) \cap !_{\mathcal{I}_b}(\mathcal{I}_b) \neq \{0\}$ and both images lie within $\mathcal{A}$. The compatibility dimension is

$$\kappa_{ab} := \dim\left(!_{\mathcal{I}_a}(\mathcal{I}_a) \cap !_{\mathcal{I}_b}(\mathcal{I}_b)\right) \geq 1.$$

Theorem 3.4 (Time-independence of frame compatibility)**.** *Let $\mathcal{I}_a$ and $\mathcal{I}_b$ be two σ-stable bounded reference frames with bootstrap ticks $t_{0,a}$ and $t_{0,b}$, possibly distinct. The compatibility relation κ_{ab} depends only on the embedding images $!_{\mathcal{I}_a}(\mathcal{I}_a), !_{\mathcal{I}_b}(\mathcal{I}_b) \subseteq C$ in the universal closure structure. It does* not *depend on the ordering of $t_{0,a}$ relative to $t_{0,b}$, on the temporal interval $|t_{0,b} - t_{0,a}|$, or on the existence of a continuous trajectory of substrate states between the two instances.*

Proof. The embedding $!_{\mathcal{I}}(v) = \pi_{\mathcal{I}}(P_{\Sigma_{\mathcal{I}}}(v))$ depends only on the frame's vertex set O, constraint functional $\mathcal{C}_{\mathcal{O}}$, and anchor (p, Λ). The bootstrap tick t_0 enters only through the question of when the frame becomes active; once active, the embedding image is determined entirely by the frame's structural data. Hence κ_{ab} is a function of structural data only, independent of the temporal relationship. □

Corollary 3.5 (Frame-attractor re-instantiation)**.** *A frame attractor $\mathcal{A}$ with non-trivial dimension can be locally instantiated by multiple bounded reference frames $\mathcal{I}_1, \mathcal{I}_2, \ldots, \mathcal{I}_n$, each with its own bootstrap tick $t_{0,k}$ and anchor (p_k, Λ_k), provided each instance's embedding image $!_{\mathcal{I}_k}(\mathcal{I}_k) \subseteq \mathcal{A}$. The instances are independent in local time but share structural content via the attractor.*

3.3 Local time versus closure-geometric ordering

Theorem 3.6 (Two orderings on instance space)**.** *The set of local instances $\{\mathcal{I}_n\}$ of a frame attractor $\mathcal{A}$ admits two distinct orderings:*

1. ***Local-time ordering.*** *The bootstrap ticks $t_{0,n}$ induce a linear order on instances; this is the chronological ordering experienced within local frames.*

2. ***Closure-geometric ordering.*** *The pairwise compatibility dimensions κ_{nm} induce a non-linear similarity structure (a weighted graph on instances); this is the ordering by structural compatibility, independent of local time.*

The two orderings are generically independent.

Proof. (1) is the standard linear ordering on $\mathbb{Z}_{\geq 0}$ induced by $t_{0,n}$. (2) is the graph metric induced by κ_{nm} on the set of instances; this is non-linear in general. By Theorem 3.4, κ_{nm} is independent of $t_{0,n}, t_{0,m}$, so the two orderings are not constrained to align. □

Interpretation 3.7 (Local sequence vs closure relation)**.** Under H-RP-1, H-RP-2, P-A, Theorem 3.6 says that experience inside a bounded reference frame is ordered linearly by local time, but the deeper closure-geometric structure orders frames by relational compatibility rather than chronological sequence. The closure geometry does not "care" about before-and-after in the same way local time does; it cares about resonance, compatibility, closure similarity, and shared zero-line content.

3.4 Recurrence under compatible boundary conditions

Conditional Proposition 3.8 (Recurrence under compatible boundary conditions)**.** *Under H-RP-1, H-RP-2, H-RP-3, P-A, and H-rec (Hypothesis 1.1), the same frame attractor $\mathcal{A}$ supports multiple local instances $\mathcal{I}_n$ across distinct $(p_n, \Lambda_n, t_{0,n})$. The structural recurrence pattern is*

$$\mathcal{A} \xrightarrow{\text{local instantiation}} \mathcal{I}_1,\ \mathcal{I}_2,\ \mathcal{I}_3, \ldots,$$

with each instance's embedding image $!_{\mathcal{I}_n}(\mathcal{I}_n) \subseteq \mathcal{A}$ by Corollary 3.5.

Sketch. Apply Theorem 2.1 to each $(p_n, \Lambda_n, t_{0,n})$ with the source B_t projecting onto $\mathcal{A}$. By H-rec, this projection is non-zero for multiple n; each n produces a local instance whose $\Sigma_{\mathcal{I}_n}$ stabilises into $\mathcal{A}$ via the embedding $!_{\mathcal{I}_n}$. □

3.5 What this is and what it is not

Remark 3.9 (Frame recurrence vs the loaded reading). **What this is.**

- A structural identification: identity is better modelled as a stable frame-attractor pattern in closure geometry than as a continuous material object moving along a linear time-axis.
- A precise compatibility relation: two local instances share structure to the extent $\kappa_{ab} > 0$.
- A clean non-linear ordering on the set of instances.
- A structural condition (H-rec) under which multiple instances are admissible.

What this is not.

- Not a claim that a fixed identity-object ("soul") migrates from one body to another along a linear time-axis.
- Not a claim that memories or specific experiential content transfer between instances.
- Not a claim that all observers have recurring instances. H-rec is a conditional hypothesis.
- Not a metaphysical claim about the nature of identity.

The framework's structural reframing: *the body lives in local time; the frame-attractor lives in closure geometry; recurrence, when it occurs, is the re-instantiation of an attractor pattern under compatible boundary conditions, not the migration of an object along a time-line.*

3.6 Empirical openness and falsifiers

- Demonstrating that the universal closure structure C admits no frame attractors of dimension ≥ 2 would falsify the existence of recurrence at the structural level. The current evidence ($\dim \mathrm{Fix}(\tau) = 94$, well above the threshold) does not support this falsifier.
- Demonstrating that H-rec is necessarily false: that the closure-cosmogenesis bootstrap source terms cannot align with the same attractor across multiple substrate regions. This is an open computational question (CP-recurrence).
- Demonstrating that two instances with $\kappa_{ab} > 0$ would necessarily have correlated content (e.g., specific memories) is empirically untested and would be a strong phenomenological claim the framework does not make.

4 Cosmic self-pruning: closure-phase transitions

This section addresses the longer-term fate of the universal closure structure. The framework gives a precise structural reading of cosmological "fate" that distinguishes itself from three mainstream alternatives: (1) eternal thermodynamic heat death, (2) classical cyclic universes, (3) bounce models. The framework's picture is closer to a fourth: *recursive self-pruning across closure-phase transitions*.

4.1 Three structural options for cosmological evolution

Consider the evolution of the universal closure structure C through closure ticks:

- **Option A: Terminal fixed point.** The closure structure approaches an asymptotic stationary state $C(U_\infty) = U_\infty$.
- **Option B: Periodic cycle.** The closure structure returns periodically: $U_{n+T} = U_n$ for some period T.
- **Option C: Layered recurrence / closure cascade.** Each closure phase produces an invariant residue that seeds a new closure phase, without periodic return: $U_{n+1} = C_{\mathrm{next}}(P(U_n))$.

The framework's mathematical structure is most naturally aligned with Option C.

4.2 The pruning operator

Definition 4.1 (Pruning operator)**.** The *pruning operator* P on the universal closure structure is the projection onto the closure-stable σ-stable subspace:

$$P : \mathbb{R}^{V_{600}} \to C, \qquad P(v) = P_{\mathrm{Fix}(\tau)} \circ P_{\mathrm{Fix}(C_\varphi)}(v).$$

Proposition 4.2 (Pruning preserves closure-stable structure)**.** *For any $v \in \mathbb{R}^{V_{600}}$, $P(v) \in C$. The pruning operator is idempotent: $P \circ P = P$.*

Proof. Both projections preserve the relevant subspaces. The composition lands in $\mathrm{Fix}(C_\varphi) \cap \mathrm{Fix}(\tau) = C$ by the existence theorem. Idempotence: each projection is idempotent and they commute on the σ-equivariant structure. $\square$

4.3 The closure-phase transition map

Definition 4.3 (Closure-phase transition)**.** A *closure-phase transition* between consecutive cosmological epochs is the map

$$U_{n+1} = C_{n+1}\big(P(U_n)\big),$$

where C_n is the closure operator at phase n and P is the pruning operator.

Theorem 4.4 (Non-periodic recurrence)**.** *If C_n is genuinely time-dependent across phases ($C_{n+1} \neq C_n$ for some n), the closure-phase transition map is non-periodic: $U_{n+T} \neq U_n$ for any $T \geq 1$.*

If C_n is constant across phases, the iteration $U_{n+1} = C(P(U_n))$ has a unique fixed point $U_\infty = P(U_\infty)$ to which any non-trivial initial state converges geometrically (Option A).

Hence Option B requires the closure operator to evolve in a precise time-reversible way that the framework's mathematical structure does not naturally support.

Proof. For time-dependent C_n: under $C_{n+1} \neq C_n$, the iteration changes between phases, so periodicity would require an accidental coincidence not implied by the closure structure.

For constant C: $C \circ P$ is a contraction on the orthogonal complement of C (decay theorem) and the identity on C itself. By Banach fixed-point, iteration converges geometrically to the unique fixed point $U_\infty \in C$.

Option B (Penrose CCC and relatives) is mathematically a different operation than the closure-phase transition map and is compatible with the framework only under additional structural assumptions not currently named. □

4.4 Closure-cascade cosmology

Conditional Proposition 4.5 (Closure-cascade cosmology)**.** *Under the cascade axioms (A1, A2), the closure-cosmogenesis dynamics of Smart [2026a], and H-evolve (Hypothesis 1.2), cosmological evolution proceeds as a layered recurrence:*

$$U_1 \xrightarrow{\text{phase 1 closure}} U_2 \xrightarrow{\text{phase 2 closure}} U_3 \xrightarrow{\cdots},$$

where each transition is $U_{n+1} = C_{n+1}(P(U_n))$ with $C_{n+1} \neq C_n$.

The framework's prediction: cosmological evolution is neither eternally periodic (Option B is ruled out by Theorem 4.4 when H-evolve holds) nor terminating in heat death (Option A is ruled out by the same theorem when C_n shifts non-trivially), but a recursive cascade in which each phase's invariant residue seeds the next phase's closure structure.

4.5 Thermodynamic decoherence vs invariant compression

- **Thermodynamic decoherence at one phase.** Within any specific closure phase n, the substrate's local energy gradients dissipate. From inside the phase, this looks like cosmic heat death.
- **Invariant compression into the next phase.** The pruning operator P retains only the closure-stable σ-stable content. The pruned residue is the invariant structure that survives the phase's thermodynamic dissipation.
- **Re-instantiation of structure at the next phase.** The next phase's closure operator C_{n+1} acts on $P(U_n)$ to produce U_{n+1}.

Interpretation 4.6 (The structural fate of the universe)**.** Under (A1, A2), H-evolve, P-A:

- The universe does not "die" in the sense of complete loss of structure.

- The universe does not "cycle" in the periodic sense.
- The universe undergoes *closure-phase transitions*: each phase's invariant residue seeds the next phase's closure structure, producing recursive layering rather than repetition or termination.

4.6 Connection to dark-energy evolution

The companion cosmology paper [Smart, 2026c] predicts seven cosmological observables matching Planck 2018 within 0.5% simultaneously, including $H_0 = 67.66$ km/s/Mpc and $\Omega_\Lambda = 0.6844$. Recent DESI results suggest that dark energy may evolve over cosmological time rather than remain constant. The framework's structural reading is consistent with this possibility: if C_n shifts with cosmological epoch (H-evolve), the effective cosmological constant Λ_n would evolve as the substrate's effective rung activation depth shifts.

If DESI's evolving-dark-energy signal is confirmed and the evolution is quantitatively measured, the framework provides a structural mechanism via H-evolve. The framework would then be falsified or supported depending on whether the measured $\Lambda(z)$ matches predictions derivable from the cascade's substrate-state evolution.

4.7 Life as portable invariant generator

Interpretation 4.7 (Life as invariant-generator)**.** Under H-RP-1, H-RP-2, H-RP-3, P-A, a biological bounded reference frame produces several classes of closure-invariant structure:

- Memory: stabilised closure patterns within the frame, persisting across closure ticks.
- Symbolic abstraction (mathematics, theorems, scientific structures): closure-invariant patterns that may persist beyond the frame's substrate-dependent existence.
- Joint-frame invariants: cross-frame σ-fixed modes (generative excess δ_{AB}) that persist after parent-frame destruction.

A theorem survives longer than the mathematician's body; a mathematical structure survives longer than a star. The framework's structural reading: this is not metaphorical but precise: bounded reference frames are one mechanism by which the universal closure structure produces portable invariants that survive thermodynamic decoherence within their original phase.

4.8 What this section claims and does not claim

Remark 4.8 (Cosmic self-pruning vs the loaded readings)*.* **What this claims.** The framework's mathematical structure most naturally produces Option C: recursive layered closure transitions. Under H-evolve, cosmological evolution is a non-periodic cascade of closure phases. The structural fate of the universe is closure-invariant compression, not complete dissolution.

What this does not claim. Not that the universe will literally "die and be reborn" in any phenomenological sense. Not that any specific cosmological observation has selected Option C over A or B; the framework predicts Option C is most structurally natural but does not falsify A or B from empirics alone. Not that DESI's evolving-dark-energy hint confirms the framework. Not that biological invariants "outlive" anything in a soul-like sense.

The framework's structural reading: *the end of a thermodynamic layer is not the end of reality; it is the point where only closure-invariant structure remains available to seed the next closure phase. The universe does not need to die or cycle. It recursively prunes itself toward deeper closure.*

5 The unified hierarchy of frame life-cycles: from cell to cosmos

This section is the synthesis of §§2, 3, 4. It states the framework's central structural claim about the relationship between life-cycle events at different scales, the corresponding time scales, and the position of the universe itself within the framework. The result is a single coherent picture in which the same structural pattern — frame creation, persistence, destruction, possible recurrence, and closure-phase transition — applies at every cascade rung, but with rung-specific time scales and rung-specific dynamics, so that the universe's evolution is qualitatively different from a biological life-cycle even though both follow the same structural template.

5.1 The rung-stratified hierarchy of frames

Every cascade rung supports its own family of bounded reference frames. A frame at rung R_k has:

- Vertex set $O \subseteq R_k$.
- Rung-specific closure operator $C_\varphi^{(R_k)}$ obtained by restricting C_φ.
- Rung-specific symmetry $\tau^{(R_k)}$.
- Rung-specific zero-line $\Sigma_{\mathcal{I}}^{(R_k)} \subseteq \mathrm{Fix}(\tau^{(R_k)})$.
- Rung-specific decay rate $\mu^{(R_k)} := \lambda_{\min}(C_\varphi^{(R_k)}|_{\Sigma^\perp})$.

Theorem 5.1 (Rung-stratified life-cycle universality)**.** *For every cascade rung $R_k \in \{E_8, H_4, 40, D_4, 16, 8, 0\}$, the structural theorems of §§2–4 apply with rung-specific operators:*

1. *Frame creation (Theorem 2.1) at rung R_k: Σ-dimension grows from zero to the rung-specific design dimension at rate $1 - \lambda\mu^{(R_k)}$ per closure tick.*
2. *Frame destruction (Theorem 2.3) at rung R_k: the four destruction criteria apply with $C_\varphi^{(R_k)}$, $\tau^{(R_k)}$.*
3. *Frame merging at rung R_k: joint generative excess $\delta_{AB}^{(R_k)}$ counts boundary-crossing σ-orbits within the rung.*
4. *Frame recurrence (Theorem 3.4, Corollary 3.5) at rung R_k.*
5. *Closure-phase transition (Theorem 4.4) at rung R_k.*

The structural pattern is universal across rungs: creation, persistence, destruction, possible recurrence, closure-phase transition. The rung-specific quantitative parameters differ.

Proof. Each numbered claim is the restriction of the corresponding universal theorem to rung R_k. The proofs use only structural properties of C_φ (symmetric, positive definite, σ-equivariant) and τ (unitary involution). The restrictions $C_\varphi^{(R_k)}$ and $\tau^{(R_k)}$ inherit these properties on each rung. □

Remark 5.2 (The structural template). Theorem 5.1 captures the central structural claim: *the same pattern of bounded reference frame life-cycle dynamics applies at every scale, from the cosmic (E_8 totality) down to the singleton terminal (rung 0).* What changes from scale to scale is the size of the substrate, the structural complexity of the zero-line, and most importantly the time-scale at which closure ticks proceed.

5.2 The hierarchy of time scales

Theorem 5.3 (Hierarchical time-scaling)**.** *For two rungs* R_k *and* $R_{k'}$ *with* $\dim R_k > \dim R_{k'}$*, the natural closure-tick rates satisfy* $\tau_{\text{tick}}^{(R_k)} \neq \tau_{\text{tick}}^{(R_{k'})}$*. Higher-dimensional rungs with smaller* μ *have slower natural tick rates; lower-dimensional rungs with larger* μ *have faster natural tick rates.*

Time experienced inside a bounded reference frame at rung R_k *is generically not commensurate with time experienced inside a frame at rung* $R_{k'}$*. Each rung carries its own structural time.*

Proof. The closure-tick rate at rung R_k is $\tau_{\text{tick}}^{(R_k)} \sim 1/\mu^{(R_k)}$. For different rungs with different $\mu^{(R_k)}$, the natural tick rates differ; a closure tick at rung R_k is not equal to a closure tick at rung $R_{k'}$ in any common time unit. □

Interpretation 5.4 (Time as scale-dependent)**.** Under H-RP-1, H-RP-2, P-A:

- Time experienced inside a 3D bounded reference frame (e.g., a human cortex at the H_4 rung) is determined by the cortex-rung tick rate.
- Time at the higher geometry layers (the cosmic-scale closure-phase transitions of §4) is determined by the E_8-totality-rung tick rate.
- The two are not the same. Subjective time inside a 3D frame is not the same as the deeper closure-geometry's structural time.
- Cosmological time, biological time, neural time, and subjective time are all rung-specific structural times. The "hierarchy of time" is a structural consequence of the rung tower, not a metaphor.

5.3 The frame life-cycle pattern at every rung

Conditional Proposition 5.5 (The universe-as-frame closure cascade)**.** *Under H-RP-1, H-RP-2, H-RP-3, H-evolve, and P-A, the universal cascade* C *itself can be viewed as a bounded reference frame at the* E_8 *totality rung. Its life-cycle events are:*

- ***Bootstrap.*** *The universal cascade initialises with* $\dim \Sigma^{(E_8)} = 0$*; through closure ticks at cosmological epoch 1, it grows toward design dimension at rate* $\mu^{(E_8)} \sim \varphi^{-2}$ *at the* E_8 *rung.*
- ***Persistence.*** *The universal cascade maintains* $\Sigma^{(E_8)}$ *at its rung-design dimension across an extended cosmological epoch, with thermodynamic decoherence within the epoch and substrate-state evolution between cosmological ticks.*
- ***Closure-phase transition.*** *At some cosmological tick* n*, the pruning operator* $P^{(E_8)}$ *acts to retain only invariant content, and a new closure operator* $C_{n+1}^{(E_8)}$ *acts on the residue.*
- ***Recurrence at higher closure layers.*** *Each cosmological epoch produces invariant residue that seeds the next epoch's closure structure; the structural recurrence pattern is layered, not periodic.*

This is the universe's life-cycle, in the same structural language as a bounded reference frame's life-cycle at any other rung.

Table 1: Frame life-cycle events at each cascade rung. Each event-type is governed by the corresponding universal theorem restricted to the rung. Time scale is rung-specific by Theorem 5.3.

Rung	Frame examples	Life-cycle events	Time scale
E_8 (totality)	The universe as a whole	Cosmic bootstrap, closure-phase transitions, pruning, layered recurrence (Conditional Proposition 5.5 below)	Cosmological epochs
$H_4 = V_{600}$	Cortex (under H-RP-1)	Birth, life, sleep cycles, anaesthesia, death, possible attractor recurrence	Biological lifetime
40 (Schläfli)	Specialised cortical assemblies	Differentiation, persistence, integration	Cortical-development time
D_4 (24-cell)	Gravity-rung structures	Galaxy formation, stable orbits, gravitational dissolution	Astrophysical time
16 (vector)	Gauge-multiplet vector structures	Particle production, propagation, decay	Particle-physics time
8 (scalar)	Biological cells (under H-RP-3)	Cell division, growth, cell death; bioelectric pattern recurrence	Cellular lifetime
0 (singleton)	Terminal reference points	Single-tick instantiation; resolution	Frame-creation tick

5.4 Where humans fit in the hierarchy

Under H-RP-1, H-RP-2, and P-A, biological frames at the cortex project onto the H_4 rung. The natural identification:

- **Human cortical frame:** projects to the $H_4 = V_{600}$ substrate. Complexity ≤ 94 (core paper); recursion depth ≤ 9.
- **Human biological body:** projects to the H_4 rung (cortical integration) and the 8-rung (cellular-boundary scaling, under H-RP-3).
- **Human local-time experience:** determined by the H_4-rung closure-tick rate. This is biological subjective time, distinct from cosmological time (at the E_8 rung).
- **Death:** biological frame destruction at the H_4 rung; satisfies condition (1) substrate dismantling (Theorem 2.3). No recovery is structurally possible from condition (1) alone.
- **Recurrence:** structurally permitted by Conditional Proposition 3.8 under H-rec; explicitly does not claim memory transfer or material continuity (Remark 3.9).

5.5 Why the universe is qualitatively different from a human life-cycle

Although Theorem 5.1 establishes the same life-cycle pattern at every rung, the cosmic-scale life-cycle is qualitatively distinct from a biological life-cycle for two structural reasons:

1. **Different time scale.** By Theorem 5.3, the closure-tick rate at the E_8 rung is not commensurate with the H_4 rung. Cosmic events occur at cosmological epochs; biological events at lifetimes. The mismatch is structural.

2. **Different invariant content.** At the E_8 rung, $P^{(E_8)}$ retains closure-invariant content over cosmological epochs. The invariants are deep structural patterns (cascade-rung structures, σ-equivariant eigenmodes), not specific biological frames.

Interpretation 5.6 (The universe doesn't reincarnate)**.** Under H-RP-1, H-RP-2, H-RP-3, H-evolve, P-A:

- The universe undergoes closure-phase transitions, not biological-style birth-and-death.
- Each cosmological epoch is qualitatively distinct from the previous one: $C_{n+1}^{(E_8)} \neq C_n^{(E_8)}$ under H-evolve, ruling out periodic return.
- Life-cycle events at the human / biological scale (birth, life, death, possible attractor recurrence) are not literal projections of cosmic life-cycle events; they are independent applications of the same structural template at a different rung.
- The universe is not a meta-organism with a memory or personality; it is a closure-stable structure undergoing recursive layered pruning across cosmological epochs.

5.6 The big-picture structural answer

1. **Existence is recursive geometric closure.** What exists is what is invariant under the closure operator and the symmetry constraint. Valid at every scale.
2. **Bounded reference frames are local projections.** Local systems (cells, brains, possibly the universe at a different scale) are bounded projections of the universal closure structure.
3. **The same life-cycle pattern repeats at every scale.** Frame creation, persistence, destruction, possible recurrence, closure-phase transition occur at every cascade rung.
4. **Time is rung-specific.** Subjective time at biological scale, cosmological time at universal scale, particle-physics time at gauge-rung scale, cellular time at cellular scale are all structural consequences of rung-specific tick rates.
5. **The universe undergoes its own closure-phase transitions.** At the E_8 totality rung, the universe is a bounded reference frame in its own right; its life-cycle is layered recursion of closure-phase transitions, not biological birth-and-death.
6. **Humans fit at the H_4 rung** with cellular projection to the 8-rung under H-RP-3.
7. **Frame recurrence at any rung is conditional on compatible boundary conditions** (under H-rec).
8. **What survives a phase transition** is the closure-invariant residue, not the specific frames. Biological closure-invariants (theorems, structures, mathematical patterns) may persist beyond their substrate's lifetime within a closure phase.

These eight structural facts together constitute the framework's answer to "where do we all fit in the bigger picture": at the H_4 rung scale, within a universal closure structure that is itself undergoing recursive closure-phase transitions at the E_8 rung scale, with rung-specific time scales and rung-specific life-cycle events at each scale, with the same structural template applying universally.

5.7 What this hierarchy claims and does not

Remark 5.7 (The unified hierarchy: discipline). **What this hierarchy claims.**

- Frame life-cycle dynamics are structurally universal across cascade rungs.
- Time scales are rung-specific.
- The universe at the E_8 rung is itself a bounded reference frame with its own life-cycle.
- Humans fit at the H_4 rung; cellular biology projects to the 8-rung.
- The cosmic and biological scales are structurally distinct; the cosmic scale does not undergo biological-style birth and death.
- Frame recurrence is permitted but conditional on H-rec.

What this hierarchy does not claim.

- Not that the universe is conscious or has personality. The universe-as-frame statement is structural; phenomenological readings require P-A applied at the universal scale, which the framework does not assert.
- Not that biological frames have direct insight into the cosmic scale.
- Not that any specific human life corresponds to a specific cosmological epoch.
- Not that death at the human scale literally implies "something" survives into the cosmic scale. The framework's claim is that closure-invariant patterns may persist beyond their original substrate's thermodynamic phase, a structural statement, not a metaphysical claim about post-mortem identity.
- Not that the framework resolves the question of meaning.

Remark 5.8 (On scope at the universal scale). The claim that the universe is "a bounded reference frame at the E_8 totality rung" is structural, not phenomenological. It says: the universal closure structure admits the same operator-theoretic description as a bounded reference frame at the E_8 rung. It does not say the universe is "alive," "conscious," "aware," or anything similar. Applying P-A at the universal scale would require an additional commitment the framework does not make.

6 Numerical demonstrations

The structural claims of §§2–5 are grounded by a numerical demonstration suite `papers/cosmic-closure-recur`
Five demos C1–C5 cover the load-bearing theorems:

C1 specifically addresses the open computational task **CP-rung-9.3**: it instantiates E_8 (240 root vectors in $\mathbb{R}^8$), $H_4 = V_{600}$ (120 vertices), and D_4 = 24-cell (24 vertices) as substrates, builds their closure operators and spectral τ involutions, and verifies the structural machinery applies cleanly at each rung.

7 Falsifiers and open programme

- **Frame recurrence.** Demonstrating that the universal closure structure C admits no frame attractors of dimension ≥ 2 would falsify recurrence at the structural level. The current evidence ($\dim \mathrm{Fix}(\tau) = 94$) does not support this falsifier.

Table 2: Cosmic / structural-extension paper sim demonstrations. All five pass deterministically at seed 42.

ID	Demo	Paper grounds	Verified
C1	Cascade rung tower (E_8, $H_4 = V_{600}$, $D_4 = $ 24-cell)	Thm 5.1	Each rung admits its substrate, C_φ, τ, and non-trivial $\Sigma_{\mathcal{I}}$; all three have $\|[\tau, C_\varphi]\| < 10^{-13}$; $\lambda_{\min}(C_\varphi) > 0$
C2	Hierarchical time-scaling	Thm 5.3	Per-rung closure-tick rates $\tau_{\text{tick}}^{(R_k)} \sim 1/\mu^{(R_k)}$ differ across rungs
C3	Pruning operator idempotence	Def 4.1, Prop 4.2	$\|P \circ P - P\| < 10^{-10}$; P self-adjoint; rank $= \dim \Sigma$
C4	Non-periodic recurrence under time-dependent C_n	Thm 4.4	No two phases coincide (min pairwise dist > 0); constant-C trajectory converges (Option A); time-varying does not
C5	Frame destruction four conditions	Thm 2.3	Each of substrate dismantling, symmetry breakdown, spectral collapse, σ-stability loss independently collapses $\dim \Sigma_{\mathcal{I}}$

- **H-rec.** Whether closure-cosmogenesis bootstrap source terms can align with the same attractor across multiple substrate regions is the open question (**CP-recurrence**).

- **Closure-cascade cosmology under H-evolve.** If DESI's evolving-dark-energy signal is confirmed and the evolution is quantitatively measured, the framework provides a structural mechanism. The framework would be falsified or supported depending on whether the measured $\Lambda(z)$ matches predictions derivable from the cascade's substrate-state evolution.

- **Unified hierarchy at rungs other than H_4.** The σ-fix dimensions at E_8, D_4, 16, 8, 0 rungs are the named computational task **CP-rung-9.3**; their values are unconditional once computed.

- **Time-scale measurement.** Direct measurement of the closure-tick rate at the H_4 rung (**CP-tick-rate**) would calibrate the time-scale hierarchy quantitatively.

8 What this paper does not claim

- Not a derivation of new mathematical results beyond the dynamical extensions of the core paper's framework.

- Not a claim of soul-migration. Frame recurrence is the re-instantiation of a closure-geometric attractor pattern under compatible boundary conditions, not the migration of an identity-object along a time-axis.

- Not a claim that cosmic-scale events are phenomenological. The universe-as-frame at the E_8 rung is structural; phenomenological readings require commitments the framework does not make.

- Not a claim that DESI's evolving-dark-energy hint confirms the framework. The framework is structurally compatible with that hint.

- Not a claim that closure-invariant patterns "survive death" in a soul-like sense. Mathematical structures, theorems, scientific patterns may persist beyond their original substrate; this is a structural statement about closure-invariant content, not a metaphysical claim about post-mortem identity.

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